

Total No. of Questions : 6]

SEAT No. :

P1390

[Total No. of Pages : 2

BE/ Insem/APR.- 18
B. E. (Mechanical S/W)
MECHANICAL SYSTEM DESIGN
(2012 Course) (402048)

Time : 1½ Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q4, Q.5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Which are the preferred number series used in the design of a multispeed gear box for geometric progression ratio? **[4]**

b) State and explain the parameters used in kinematic design of gearbox. **[6]**

OR

Q2) Draw the structure and gear box diagrams for the following equations of twelve speed gear box and determine the maximum transmission range for each equation for R 5 Series. $\phi = \sqrt[5]{10}$ **[10]**

a) $Z = 2(6) 2(1) 3(2)$

b) $Z = 2(3) 2(6) 3(1)$

c) $Z = 2(6) 2(3) 3(1)$

Q3) a) Explain design and natural tolerances **[4]**

b) Define the following terms: **[6]**

i) Population

ii) Sample

iii) Random variables

Also explain the concept of reliability based design.

OR

P.T.O.

- Q4)** A particular type of rolling contact bearing has a normally distributed time to failure with a mean of 10000 hours and a standard deviation of 750h. If there are 100 such bearings fitted at a time, how many May be expected to fail within the first 11000h? **[10]**

z	0	1	2	3	4	5	6	7	8	9
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319

- Q5) a)** Explain the steps involved in belt conveyor analysis? **[4]**
- b)** A three idler through belt, horizontal conveyor is to be used for transporting 500 ton/hr of coal having weight density 8000 N/m³. the surcharge factor 'c' for three idler through belt is 0.1. If the belt speed is 100 m/min. Select the standard belt width for conveyor belt.
- Available standard belt width is : 400, 450, 500, 650, 750, 800, 900, 1000, 1200, 1400, 1600, 1800, 2000 mm. **[6]**

OR

- Q6)** Derive the expression for frictional force between the belt and carrying run idler and between belt and return run idler. **[10]**

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